Harry Winston Sullivan

Curriculum Vitae

Department of Chemical Engineering
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Professional Summary

Chemical Engineering PhD student with a focus on computational science and engineering. My diverse background contains professional research in computer science, physics, chemical engineering, machine learning, and finance. Highly adept and creative researcher with a proven track record of swiftly conceptualizing, designing, and implementing innovative solutions to intricate engineering challenges.

Education

2024–Present **Ph.D. Chemical Engineering**, *University of Minnesota*, Minneapolis, MN, Advisors: **Prof. Sapna Sarupria** (*link*); **Prof. Ellad B. Tadmor** (*link*).
Generative modeling for materials design, including molecular crystal packing and advanced sampling methods for rare events in molecular simulations; diffusion models and geometric deep learning; statistical mechanics; non-equilibrium thermodynamics.

2024–2026 **M.S. in Data Science for Chemical Engineering & Materials Science**, *University of Minnesota*, Minneapolis, MN, GPA: 3.968; completed as the first stage of the Ph.D. program; program (*link*).
Interdisciplinary training in statistical and data analysis, machine learning, optimization, and computational applications in chemical and materials engineering.

2019–2023 **B.S. Computational Physics**, *University of Utah*, Salt Lake City, UT, GPA: 3.868.
Classical mechanics; electricity and magnetism; statistical mechanics; quantum mechanics; solid state physics; scientific computing.

2019–2023 **Minor, Computer Science**, *University of Utah*, Salt Lake City, UT.
Bayesian machine learning; uncertainty quantification; model validation; data structures and algorithms; neural networks.

Industry Experience

2026–Present **AI Intern — Generative Models for Microelectronics**, *Matterstack (link)*.
Summer internship developing generative modeling approaches for applications in microelectronics.

2023–2023 **Machine Learning Intern — Cryptocurrency Markets**, *Blank Page Studios (link)*, Remote.
Summer internship applying machine learning and data analysis to cryptocurrency markets and NFT sales analytics.

Research Experience

2023–2024 **Graduate Research Assistant — Neutron Scattering, Quantum Molecular Simulation, and Machine Learning**, *University of Utah*, Salt Lake City, UT, Advisor: **Dr. Michael Hoepfner**, Associate Professor (*link*).
Applied quantum statistical mechanics with variational optimization to infer interatomic forces from soft-matter microstructure.

2021–2023 **Undergraduate Researcher — Neutron Scattering, Molecular Simulation, and Machine Learning**, *University of Utah*, Salt Lake City, UT, Advisor: **Dr. Michael Hoepfner**, Associate Professor (*link*).
Used classical statistical mechanics and Bayesian machine learning to infer interatomic forces from soft-matter microstructure.

2021–2021 **Research Assistant — Gravitational-Wave Data Analysis (LIGO)**, *University of Utah*, Salt Lake City, UT, Advisor: **Dr. Yue Zhao**, Assistant Professor (*link*).
Summer appointment. Refactored LIGO analysis workflows in Python and Fortran and authored standard operating procedures for future analyses.

Publications

Feb 2026 **C. Zeng, H. W. Sullivan, T. Egg, M. M. Martirossyan, P. Höllmer, J. Jin, R. G. Hennig, A. Roitberg, S. Martiniani, E. B. Tadmor, M. Liu**, *MolCrystalFlow: Molecular Crystal Structure Prediction via Flow Matching (link)*, arXiv preprint.

Oct 2025 **H. W. Sullivan, B. L. Shanks, M. Cervenka, M. P. Hoepfner**, *Physics-Informed Gaussian Process Inference of Liquid Structure from Scattering Data (link)*, J. Phys. Chem. B 2025, 129(45), 11802–11815.

Apr 2025 **B. L. Shanks*, H. W. Sullivan, P. J. Jungwirth, M. P. Hoepfner**, *Experimental Evidence of Quantum Drude Oscillator Behavior in Liquids Revealed with Machine Learning Assisted Iterative Boltzmann Inversion (link)*, J. Chem. Phys. 162, 164501 (2025).

Dec 2024 **B. L. Shanks, H. W. Sullivan, M. P. Hoepfner**, *Bayesian Analysis Reveals the Key to Extracting Pair Potentials from Neutron Scattering Data (link)*, J. Phys. Chem. Lett. 2024, 15, 12608–12618.

Mar 2024 **B. L. Shanks, H. W. Sullivan, A. R. Shazed, M. P. Hoepfner**, *Accelerated Bayesian Inference for Molecular Simulations using Local Gaussian Process Surrogate Models (link)*, J. Chem. Theory Comput. 2024, 20, 3798–3808.

Contributed Talks and Posters

Jun 2026 **Riemannian Flow Models with Reinforcement Learning for Molecular Crystal Structure Prediction**, *ChemAI NYC 2026*, New York University, New York, NY, Primary presenter..

Jun 2026 **Transferable Boltzmann Sampling via Conditional Weighted Flow Matching and Energy-Based Models**, *ChemAI NYC 2026*, New York University, New York, NY, Secondary presenter with Thomas Egg..

May 2026 **Generative Machine Learning as a Surrogate for Molecular Simulation**, *Peter O. Stahl Advanced Design Forum; 2026 IPRIME Annual Meeting (AI/ML Workshop)*, University of Minnesota and Cargill, Minnesota, Same combined poster presented at both events..

Mar 2025 **Generative Modeling of Boltzmann Distributions Using Denoising Diffusion Models**, *Institute of Organic Chemistry and Biochemistry (IOCB Prague), P. Jungwirth Group*, Prague, Czech Republic.

Jul 2024 **Estimation and Uncertainty Quantification of Experimental Radial Distribution Functions Using Non-Stationary Gaussian Processes**, *Foundations of Molecular Modeling and Simulation, U.S. Department of Energy (DOE)*, Salt Lake City, UT, Poster presentation..

Jan 2024 **Current Challenges in Reconstructing Classical Force Fields from Neutron Scattering Data**, *DOE EFRC: Multi-scale Fluid–Solid Interactions in Architected and Natural Materials*, Salt Lake City, UT.

Sep 2023 **Learning Interatomic Forces from Fluid Structure with Machine-Learning-Accelerated Bayesian Optimization**, *American Chemical Society (ACS)*, Laramie, WY.

Mar 2022 **Bayesian optimized force fields enabled by a radial distribution function surrogate model**, *Recent Advances in Machine-Learning-Accelerated Molecular Dynamics, Centre Européen de Calcul Atomique et Moléculaire (CECAM)*, Trieste, Italy, Poster presentation. Presented with Brennon L. Shanks..